Group Project – Open topic

1 Datasets

Many datasets are available through public repositories, e.g.:

- The Awesome public datasets GitHub repository: https://github.com/awesomedata/awesome-public-datasets
- The Registry of Open Data on AWS - https://registry.opendata.aws/
- The Data.gov open datasets - https://www.data.gov/
- The World Bank data catalog - https://data.worldbank.org/
- The UCI Machine Learning Repository: http://archive.ics.uci.edu/ml/
- GEO (Gene expression omnibus): http://www.ncbi.nlm.nih.gov/geo/
- The Cancer Genome Atlas (TCGA): https://tcga-data.nci.nih.gov/tcga/

2 Data Munging Requirements

1. Read the data into a language-native format, properly treating NAs and other encoded variables.
2. Assess data quality via proportions or relationships of missing values, consistency in encoding, and/or numeric quality scores.
3. Recode any variables of interest as needed for exploratory analysis, which might include combining categories, aggregating summary statistics from factors, or normalizing data between categories.
4. Identify and display any relationships of interest using the guidelines below:

3 EDA Requirements

1. Produce several plots or tables of the data with formats appropriate to the data type. If possible with the data type, try to include at least five types of visualization in your EDA section.
2. These are often best if they follow a logical pattern (A suggested B, so we examined B using C), but comprehensive searches are also good.
3. Have at least one association that you examine from multiple angles (i.e., multiple types of graph or model), and interpret the combined results.

4 General requirements

- In a final paragraph, label group contributions in the style of recent journal articles ("AB produced figures, recoded variables, and wrote the summary").
- Produce a report that includes your interpretations of the results, your code, and any comments needed to interpret the purpose of the code.
- Be descriptive in linking stages of analysis. In this case, more of the emphasis is on the relationships you discover in the data than on the data processing steps.